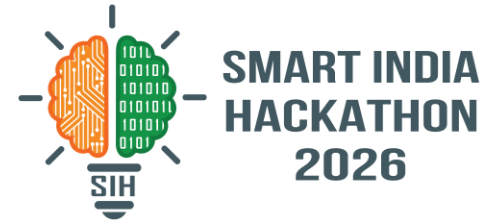
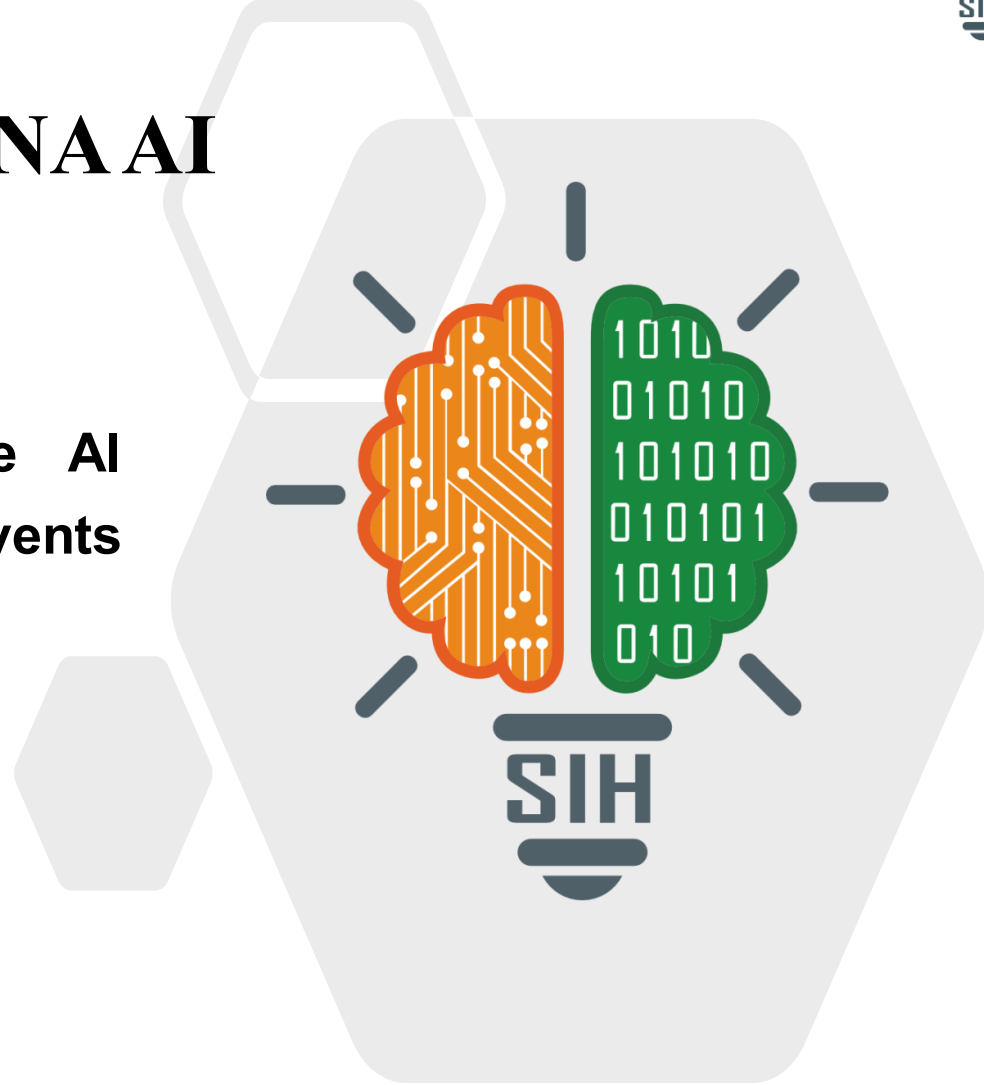


SMART INDIA HACKATHON 2026



VARUNA AI

- **Problem Statement ID – SIH260006**
- **Problem Statement Title- Explainable AI Model to Predict High-Impact Rain Events Using Satellite Data**
- **Theme- Space Technology**
- **PS Category- Software**
- **Team Name- ALL IN**





- **Problem:** Cloudbursts and extreme rain events cause floods and landslides across India; current forecasts are hard to trust or act on quickly



- **VARUNA AI:** an Explainable AI system that predicts high-impact rainfall using ISRO INSAT satellite data and weather data



- Gives a **1–72 hour** rainfall-risk forecast, classified **Low to Extreme** with a confidence score



- Explains each prediction using **SHAP and Grad-CAM** on the satellite image



- Shows risk and early-warning alerts on an interactive **India risk map** for disaster-management teams



- **Core innovation:** it doesn't just predict heavy rain — it **explains why** the AI believes the risk is high

TECHNICAL APPROACH

Explainable AI based Extreme Rainfall Intelligence & Early Warning System

1. DATA SOURCES

Satellite Data (NASA GIBS)



- Precipitation
- Cloud Cover
- Humidity
- Temperature
- Wind Speed

Weather Data (APIs)



- Forecast Data
- Historical Data
- Real-time Data

2. DATA INGESTION & PROCESSING



Data Ingestion (FastAPI Scheduler)



Raw Data Storage PostgreSQL / PostGIS (Spatial + Tabular)



Data Processing Cleaning, Resampling, Feature Engineering



Prepared Data Tabular + Raster (Analysis Ready)

3. AI & ANALYTICS LAYER



AI Models

- Tabular Models (XGBoost, Random Forest)
- Vision Models (CNN for Satellite Images)
- Hybrid Models



Prediction Engine

- Rainfall Intensity Prediction
- Risk Classification (Low / Moderate / High)



Alert Engine

- Threshold Based Alerts
- Early Warning Generation

4. EXPLAINABILITY LAYER



SHAP Analysis

- Feature Importance
- Global & Local Interpretations



Grad-CAM

- Visual Explanations for Satellite Images
- Heatmaps



Insights & Reports

- Model Insights
- Explanation Reports
- Downloadable PDFs

5. APPLICATION LAYER



Web Dashboard (React)

- Interactive Maps
- Rainfall Forecasts
- Trends & Analytics
- Alerts & Notifications



Notifications

- Email Alerts
- SMS / Push Alerts



User Roles

- Admin
- Analyst
- Public User

TECHNOLOGY STACK



Python
(Backend Development)



FastAPI
(API & Scheduler)



PostgreSQL + PostGIS
(Spatial DB)



PyTorch
(Deep Learning Models)



scikit-learn
(ML Models)



SHAP
(Explainable AI)



React
(Frontend)



Docker
(Containerization)



Leaflet
(Maps)

INFRASTRUCTURE & DEPLOYMENT



AWS Cloud
(EC2, S3, IAM)



Docker
(Containerization)



PostgreSQL + PostGIS
(Geospatial DB)



FastAPI
(Backend API)



React
(Frontend)



GitHub
(Code Repository)

Legend: —→ Data Flow

---→ Deployment / Integration

FEASIBILITY AND VIABILITY



Feasibility

- Required satellite and weather data sources are available
- Models can be developed and validated in stages
- Modular architecture allows independent testing



Challenges

- Missing or delayed data
- False alarms
- Model interpretability
- Operational decision-making



Mitigation

- Data validation and fallback sources
- Historical validation and threshold tuning
- SHAP and Grad-CAM explanations
- Human-in-the-loop decision support



Target Users

- Disaster-management authorities
- Emergency-response teams
- Weather and research teams



Benefits

- Earlier identification of high-impact rainfall risk
- Region-level risk information for faster decisions
- Better understanding of AI predictions through explanations
- Human-readable, easy-to-act-on warnings
- Prediction and alert history for auditing



- **MOSDAC / ISRO Satellite Data:** <https://www.mosdac.gov.in/>
- **NASA Earthdata:** <https://www.earthdata.nasa.gov/>
- **SHAP Documentation:** <https://shap.readthedocs.io/en/latest/>
- **Grad-CAM Paper (ICCV 2017):**
https://openaccess.thecvf.com/content_iccv_2017/html/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.html
- **Project GitHub Repository:** <https://github.com/Unnathihegde/isrosih>